# Class 17: Private Equity

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# Today's Question



**Yale’s endowment has ~40% in private equity and venture capital.**
**Are they geniuses, or just taking risks they don’t fully see?**

# Positioning



| Where We've Been | Where We Are | Where We're Going |
|---|---|---|
| Evaluating public market managers | Illiquid alternatives: PE/VC | Hedge funds |

# Learning Objectives



By the end of today's class, you should be able to:

1. Describe the PE/VC fund structure (LP/GP, carried interest, J-curve)
2. Explain why IRR can be misleading for PE performance
3. Articulate the debate over PE "alpha" vs. leverage + illiquidity premium

# Alternative Investments Introduction



Let's broaden the scope of the investable universe:

- Private equity, hedge funds, fixed income, and derivatives

Everything we learned in the first half of class applies, but...

- Data is more opaque
- Liquidity is a bigger issue
- Institutional details matter more
  - E.g. lock-up periods, redemption, fee structure
- Efficiency (maybe?)

# What Is Private Equity?



A PE fund is a financial intermediary typically formed as a limited partnership:

- **General Partner (GP):** Manages funds, makes investments
- **Limited Partners (LPs):** Passive investors (pension funds, endowments)

A PE fund invests only in private (or soon-to-be-private) companies:

- **Costs:** information asymmetry and illiquidity
- **Benefits:** control

# Private Equity: Structure



A PE fund takes an **active** role in the firm:

- Added value through management/monitoring

A PE fund has an **exit strategy:**

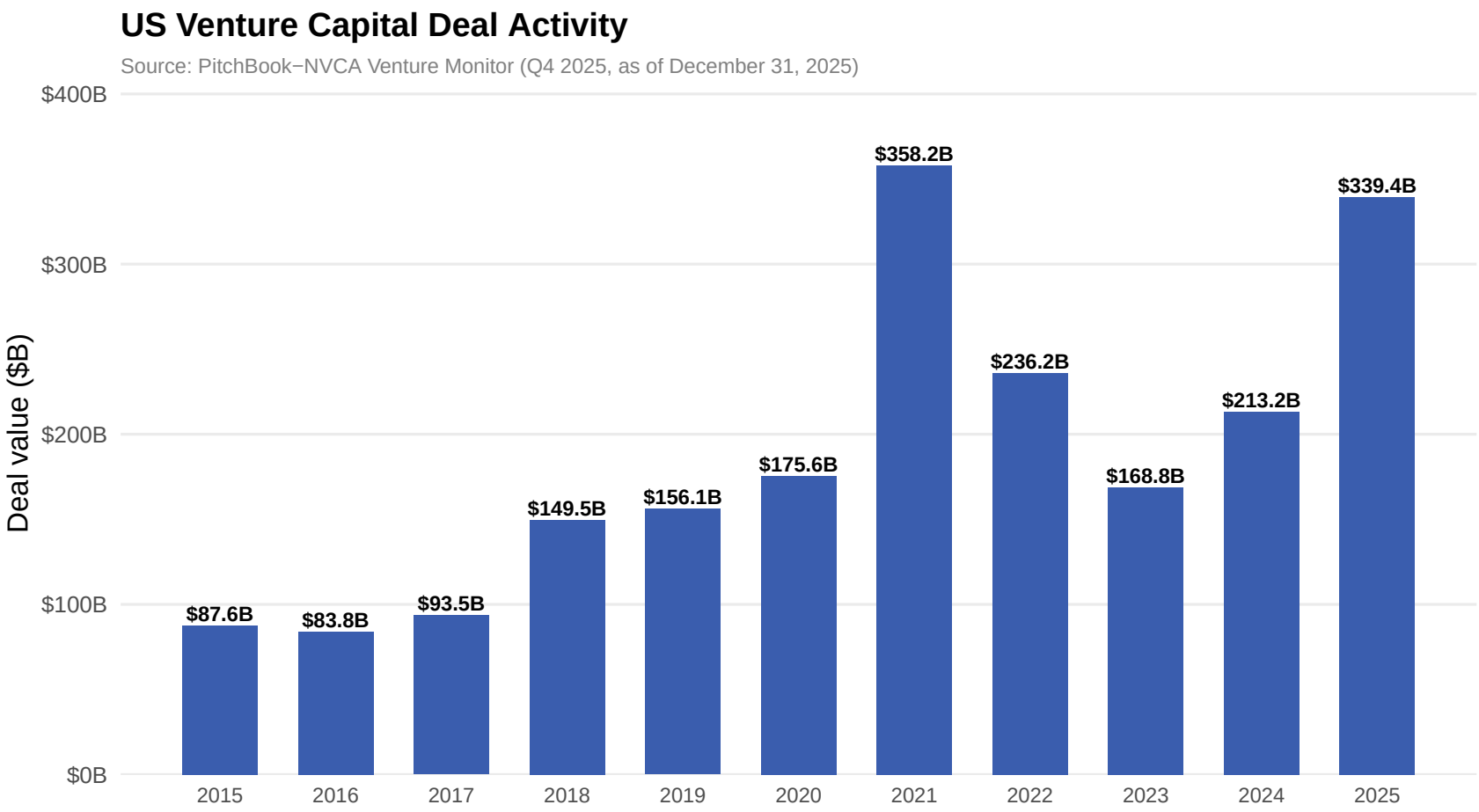
- Because of illiquidity of investment, targets must provide a clear path to exit (e.g. IPO or sale to another company)
- Funds are closed-end with lifespans of 10–12 years:
  - Commitments raised
  - Capital calls/investments made
  - “Harvesting” period

Table 2: Venture Capital-Backed Companies by CSA, 1975 - 2005

| | Portfolio Company Location | | Share of Investments in CSA | | |
|---|---|---|---|---|---|
| CSA | Share of Total | # | Main Office | Branch Office | Outside |
| San Jose/San Francisco, CA | 29.01% | 4,063 | 56.55% | 16.40% | 27.04% |
| Boston, MA | 11.67% | 1,634 | 42.34% | 8.07% | 49.59% |
| New York, NY | 8.74% | 1,224 | 47.94% | 2.37% | 49.69% |
| Los Angeles, CA | 6.08% | 851 | 11.93% | 2.53% | 85.54% |
| Washington, DC | 4.17% | 584 | 20.96% | 6.37% | 72.67% |
| San Diego, CA | 3.53% | 494 | 6.71% | 3.75% | 89.55% |
| Dallas, TX | 2.93% | 411 | 17.04% | 9.25% | 73.71% |
| Seattle, WA | 2.73% | 383 | 17.40% | 0.25% | 82.35% |
| Denver, CO | 2.63% | 369 | 22.68 | 0.55% | 76.78% |
| Atlanta, GA | 2.48% | 348 | 20.50% | 0.33% | 79.17% |
| Chicago, IL | 2.16% | 303 | 30.70% | 0.85% | 68.44% |
| Philadelphia, PA | 3.16% | 302 | 12.91% | 2.00% | 85.09% |
| Other | 21.70% | 3,040 | 16.41% | 1.19% | 82.40% |
| Total | 100% | 14,006 | 35.63% | 7.83% | 56.54% |

# Venture Capital: Deal Activity





# What Value Does Venture Capital Add?

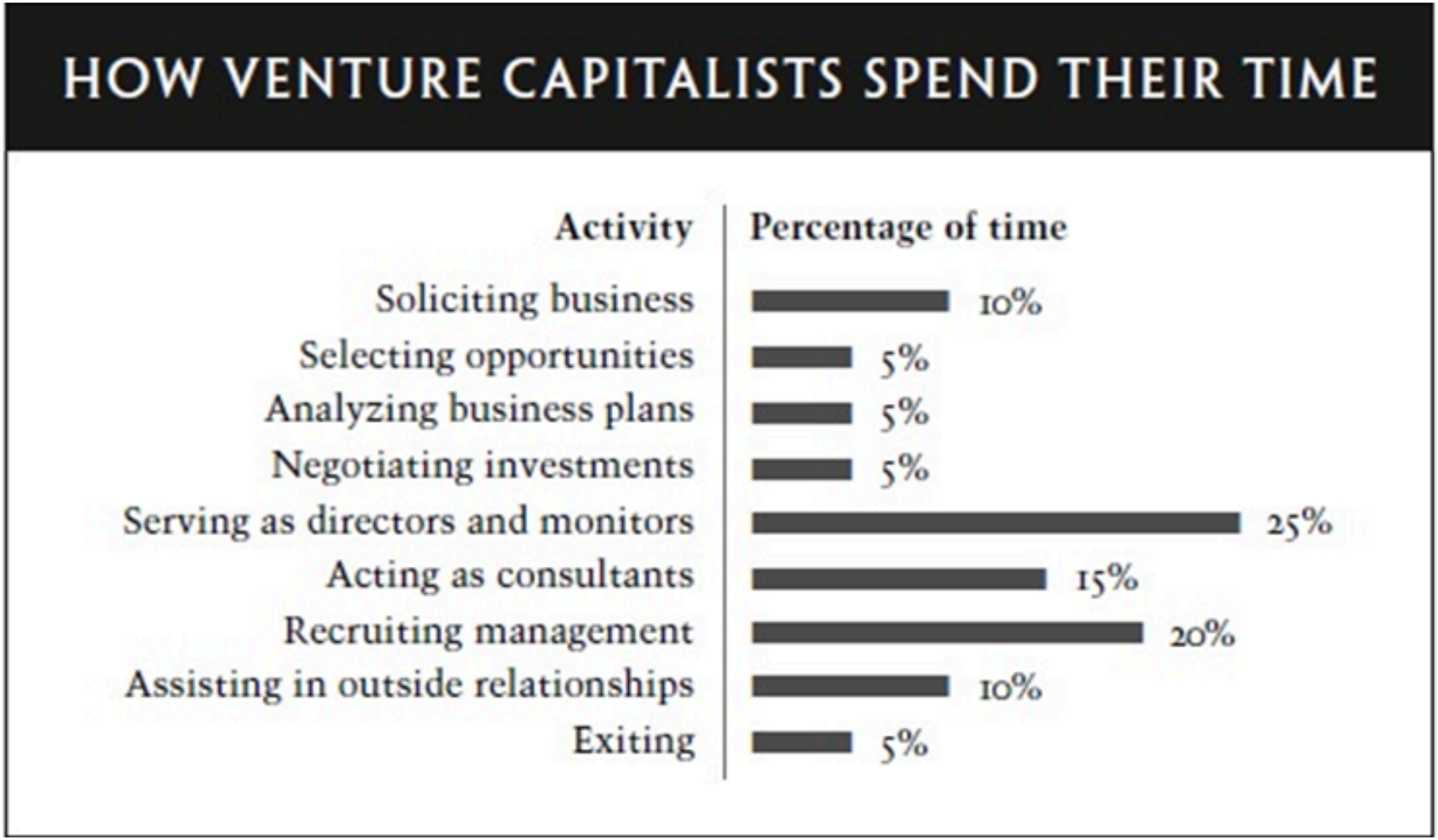


1. **Screening:**
   - VCs as winner-pickers — select 1-in-400 deals to fund
   - Do VCs bet the jockey or the horse?
   - High management turnover suggests the horse
2. **Advising/Monitoring:**
   - Requires control rights
   - Investments typically include voting rights, board seats, esp. in bad times
3. **Exit**

*Important to have good VCs for good returns!*

# How VCs Spend Their Time





# How Do Venture Capitalists Make Decisions?



Gompers et al. (2020) find that:

1. When selecting deals, VCs place the greatest importance on the **management team**
2. VCs devote substantial resources to deal sourcing, selection, and post-investment value-add
3. Most VCs **do not use DCF or NPV** techniques to evaluate investments. They rely more on valuation multiples and IRR
4. After investing, VCs are **actively involved** with their portfolio companies
5. VCs are generally inflexible on key economic terms in contracts like liquidation preferences, anti-dilution rights, and vesting
6. VC firms are small, with only 14 employees on average, half of whom are senior investment professionals. Most of their time is spent on sourcing and working with portfolio companies

# Buyout Funds



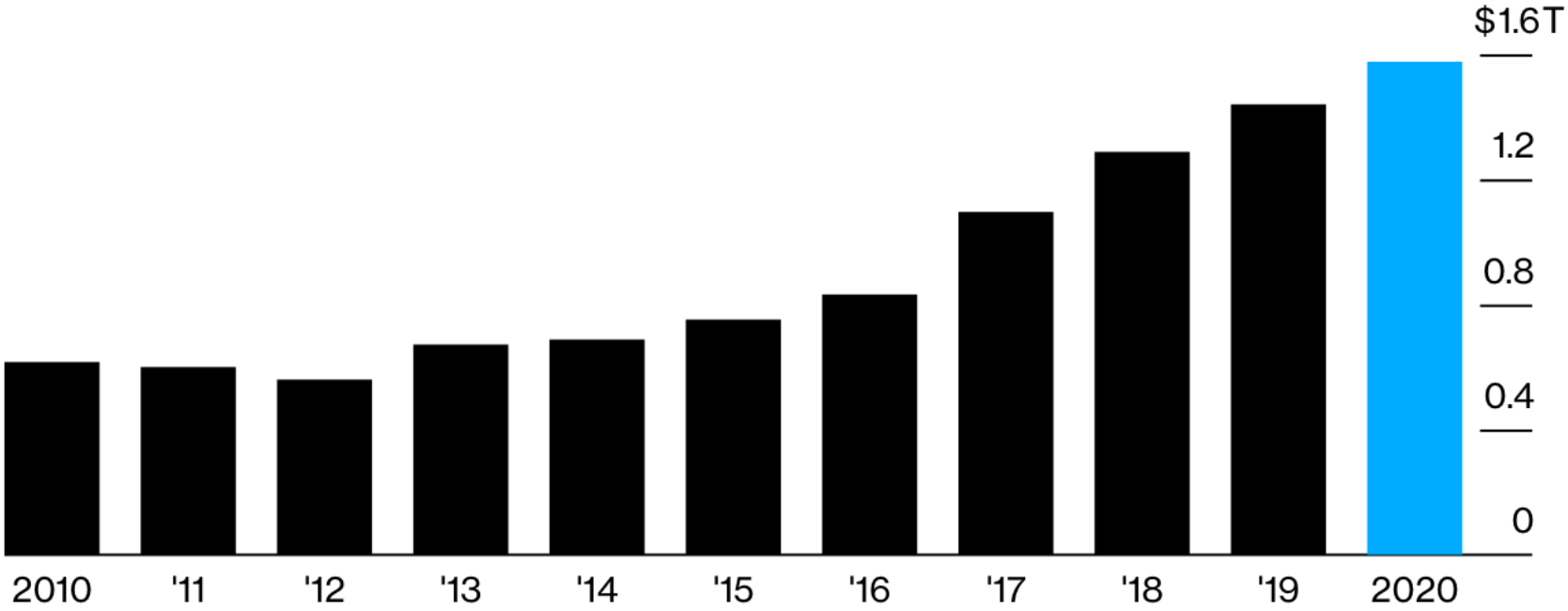
In contrast to VC, (Leveraged) Buyout (LBO) funds focus on **established firms**

- Firms are either private to begin with, or are taken private (as in a management buyout)
- Large amounts of **new debt** are used to finance the repurchase of existing equity
  - Often 90/10 debt-to-equity ratio in new company
  - Example: RJR Nabisco, Dell

There's big demand: in 2020, LBO funds had $1.6 trillion to spend on buyouts

## LBO War Chest

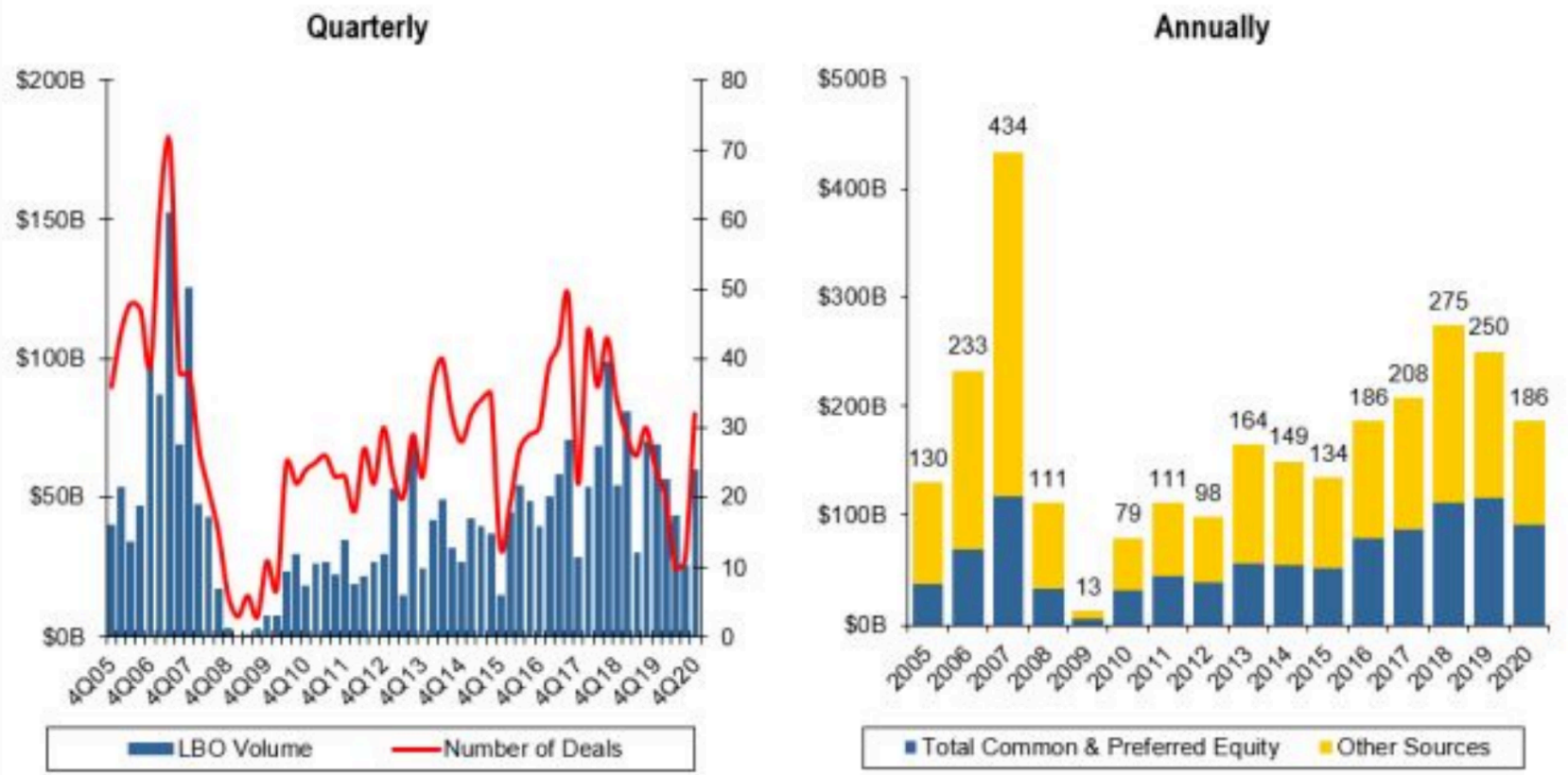
Private equity's dry powder tops $1.5 trillion




Source: Preqin
Note: 2020 as of November

# LBO Capital Structure

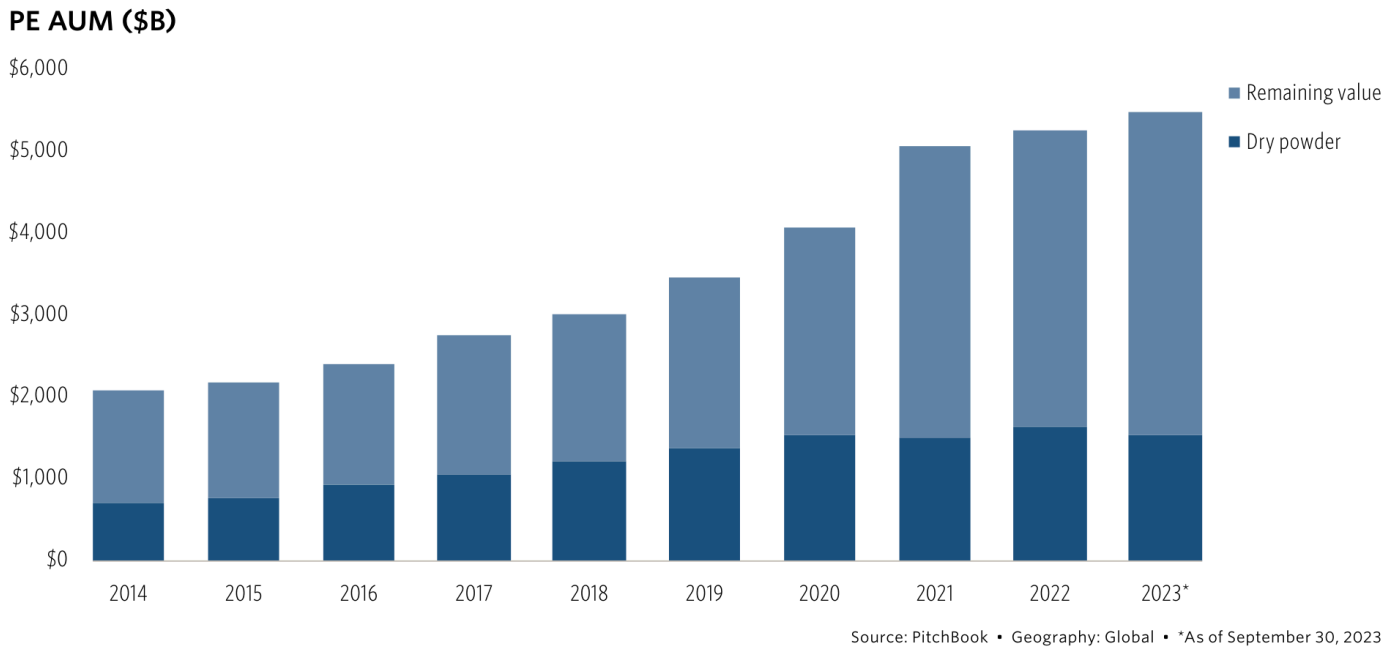





Data through Dec. 31, 2020.

# PE Assets Under Management



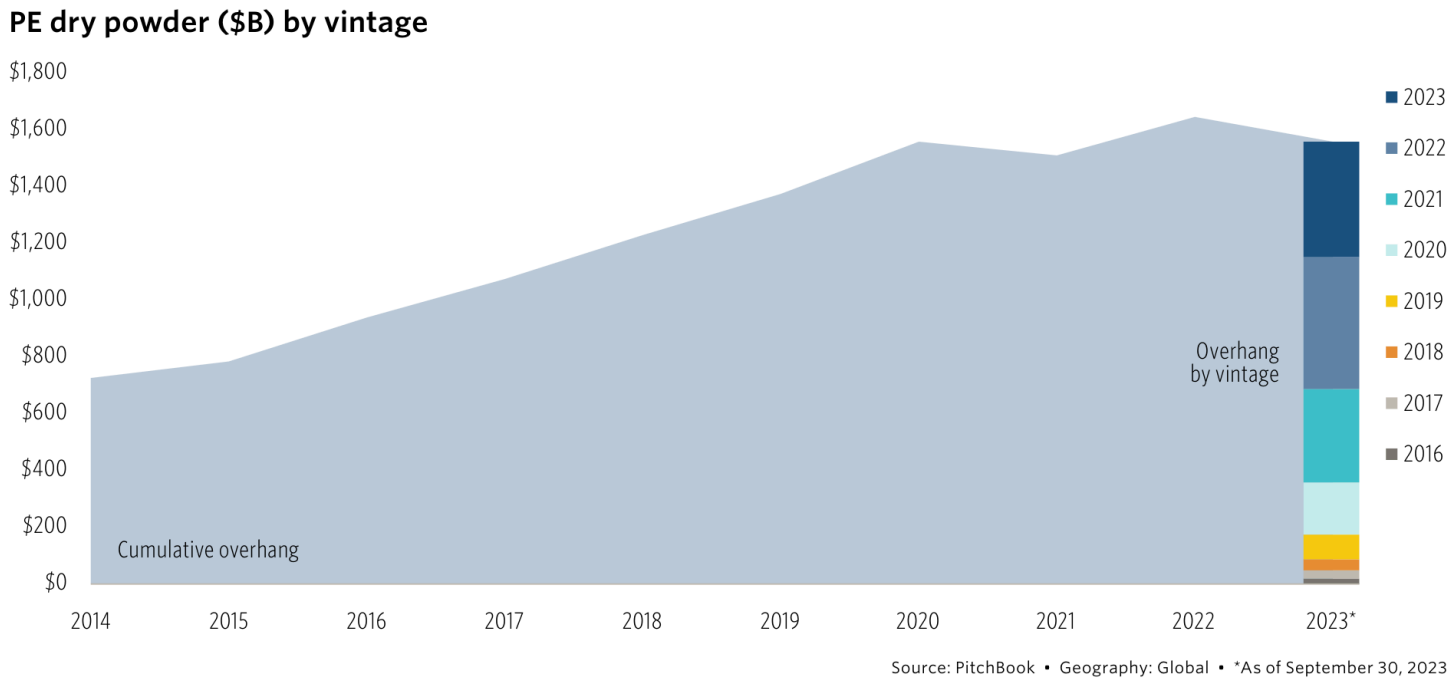



Source: PitchBook, Q1 2024 Global Private Market Fundraising Report (as of Sept. 30, 2023)

# PE Dry Powder by Vintage



PitchBook




PE AUM ($B)

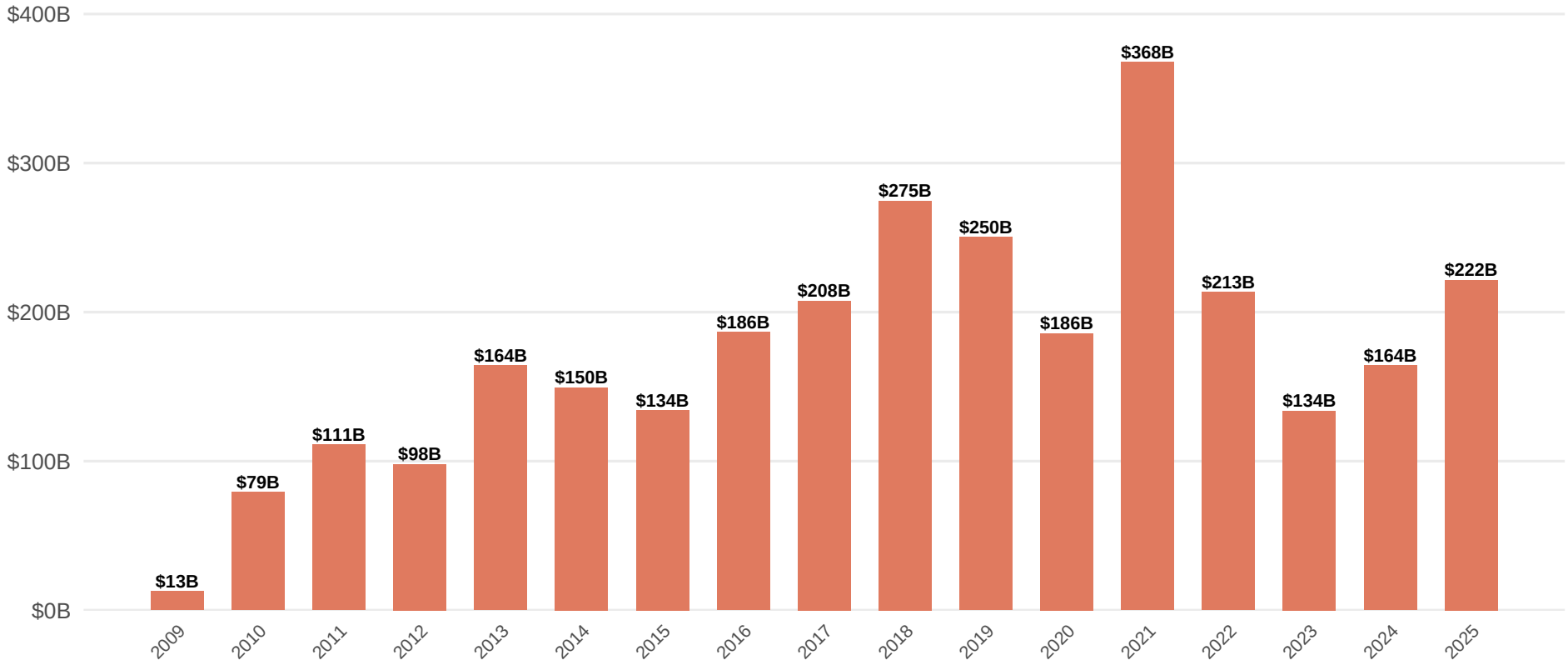
Source: PitchBook, Q1 2024 Global Private Market Fundraising Report (as of Sept. 30, 2023)

# LBO Deal Activity



## Total US LBO Transaction Volume

Includes all sources (loans, secured/unsecured debt, equity)
Source: PitchBook LCD, Q4 2025 US LBO Debt Report



# Buyout Funds: Value-Add



Where does the value-add come from?

- No need to take a company private to exploit an undervalued stock
- Similar to VC, **control must matter**

What types of things do LBO firms do once they have control?

- Spin-offs of conglomerates
  - Beatrice Foods
- Efficiency gains (cost-cutting, union negotiations, layoffs)
  - Safeway

# Or Is It Just Leverage?

# Private Equity Fund Contracts



Incentives of GPs and LPs may not always be aligned

GPs may have incentives to shirk or take excessive risk

Fees, co-investment by fund managers, and covenants discipline GPs and protect the LPs:

- **Fees:** flat rate (1–3%) plus 20–30% performance fee (carried interest)
- **Co-investment:** GPs often required to personally invest in the fund (typically 1%)
- **Covenants:** restrictions on activities

# Private Equity Fund Covenants



THE NUMBER OF COVENANTS IN 140 VENTURE PARTERNSHIP AGREEMENTS, BY YEAR

| | PERCENTAGE OF CONTRACTS WITH COVENANT IN: | | |
|---|---|---|---|
| | 1978–82 | 1983–87 | 1988–92 |
| Covenants relating to the management of the fund: | | | |
| Restrictions on size of investment in any one firm | 33.3 | 47.1 | 77.8 |
| Restrictions on use of debt by partnership | 66.7 | 72.1 | 95.6 |
| Restrictions on coinvestment by organization's earlier or later funds | 40.7 | 29.4 | 62.2 |
| Restrictions on reinvestment of partnership's capital gains | 3.7 | 17.6 | 35.6 |
| Covenants relating to the activities of the general partners: | | | |
| Restrictions on coinvestment by general partners | 81.5 | 66.2 | 77.8 |
| Restrictions on sale of partnership interests by general partners | 74.1 | 54.4 | 51.1 |
| Restrictions on fund-raising by general partners | 51.9 | 42.6 | 84.4 |
| Restrictions on other actions by general partners | 22.2 | 16.2 | 13.3 |
| Restrictions on addition of general partners | 29.6 | 35.3 | 26.7 |
| Covenants relating to the types of investment: | | | |
| Restrictions on investments in other venture funds | 3.7 | 22.1 | 62.2 |
| Restrictions on investments in public securities | 22.2 | 17.6 | 66.7 |
| Restrictions on investments in leveraged buyouts | .0 | 8.8 | 60.0 |
| Restrictions on investments in foreign securities | .0 | 7.4 | 44.4 |
| Restrictions on investments in other asset classes | 11.1 | 16.2 | 31.1 |
| Total number of partnership agreements in sample | 27 | 68 | 45 |
| Average number of covenant classes | 4.4 | 4.5 | 7.9 |
| Average number of covenant classes (weighted by fund size) | 4.4 | 4.6 | 8.4 |

# The Challenge with Evaluating Private Equity



Private equity is, by definition, private — limited trading with no continuously posted price. How can we evaluate the risk-return profile?

Two main (and limited) ways:

1. **Multiples on investment**
2. **IRR (discounted cash flows)**

Multiples: the value out divided by the initial payment in

$$M = \frac{\sum_{t=1}^{T} C_t}{P_0}$$

IRR: the rate that sets the PV of all cash flows to zero

$$0 = P_0 + \sum_{t=1}^{T} \frac{C_t}{(1+r)^t}$$

# Private Equity Performance



| | Equal Weighted | | | Size Weighted | | |
|---|---|---|---|---|---|---|
| Sample | All Funds | VC Funds | Buyout Funds | All Funds | VC Funds | Buyout Funds |
| $IRR_{VE5}$ | 0.11 | 0.10 | 0.12 | 0.13 | 0.15 | 0.12 |
| | 0.18 | 0.20 | 0.14 | 0.18 | 0.30 | 0.13 |
| | (0.36) | (0.40) | (0.24) | (0.32) | (0.46) | (0.23) |
| | [0.02;0.22] | [0.02;0.21] | [0.02;0.22] | [0.03;0.23] | [0.04;0.42] | [0.02;0.21] |

- Table reports mean, median, standard deviations, and 25th–75th percentile of dollar-weighted returns (IRR) to PE funds
  - Why dollar-weighted returns?
  - On average, roughly equivalent to S&P 500 during same period (net of fees)

# Comparing to the Market: PME



The comparable results prompt a simple comparison: what is the return of the contribution to the fund if it had been put in the market?

This is the **Private Market Equivalent (PME):** the funds placed into PE are evaluated based on the opportunity cost of missing out on the market

This approach can be much better than just IRR or multiples:

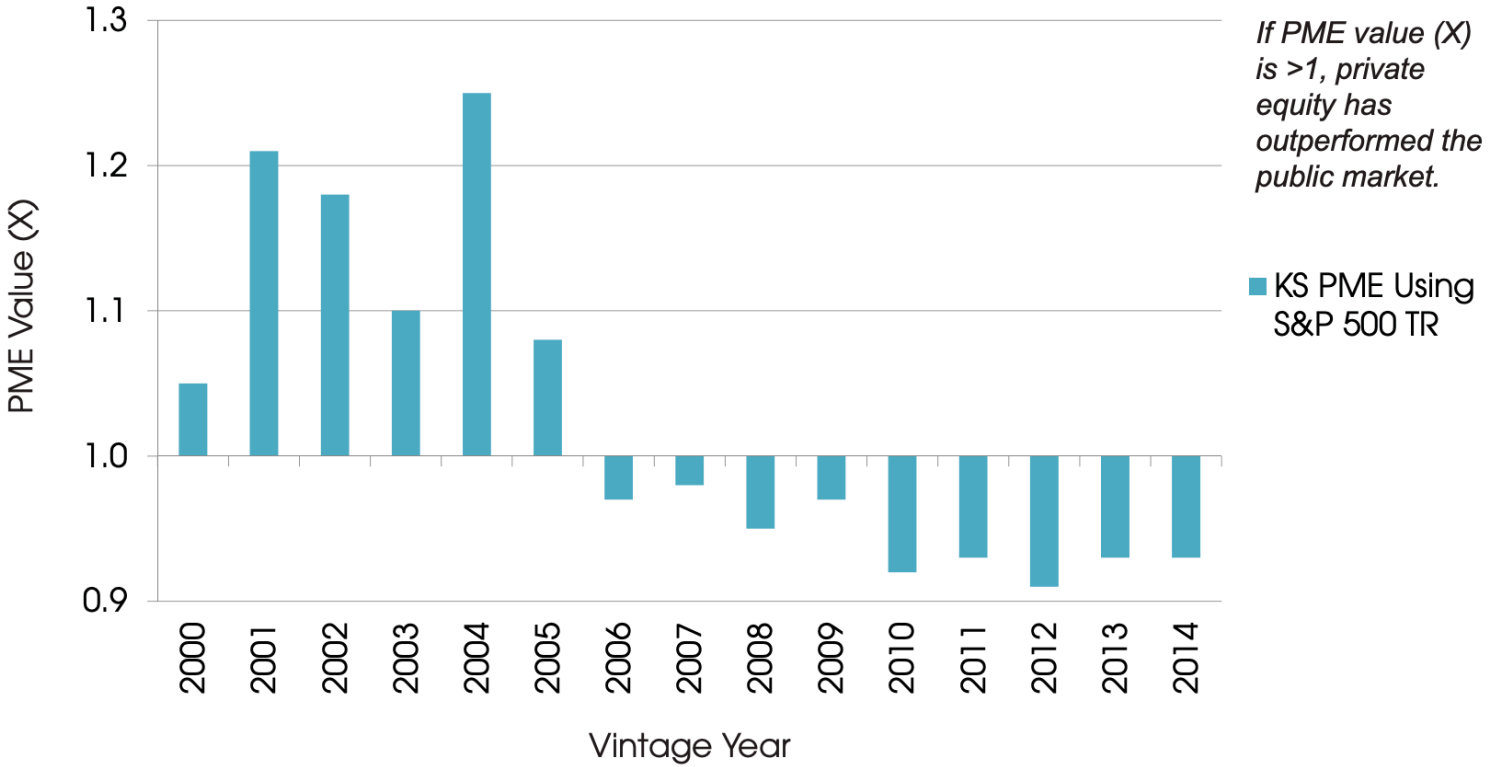
- A rising tide lifts all boats
- Effectively adjusts payouts based on the return in the market
- **But...** assumes a $\beta$ of 1!

With PME, IRR, and multiples, can compare the relative return on investments (but not riskiness)

# PE Returns by Vintage Year



**Fig. 3:** KS PME: All Private Equity - All Regions (As of 31 December 2014)




*Source: Preqin Performance Analyst*

# How to Evaluate the Riskiness of PE Investments?



PME, IRR, and multiples give ways to evaluate PE **funds**. But what about riskiness?

Two approaches:

1. Consider the variance in the time series. Subject to serious data issues (selection bias)
2. Simulations — model capital calls / cash flows

Neither of these methods adjust for aggregate risk factors.

Gupta and Van Nieuwerburgh (2020) argue that PE is actually quite exposed to other factors:

- Model the cash flows and compare to public market factors
- Highly exposed to SMB, HML, and other public equities
- *Suggestive that you could get the same return by investing in public equities instead (at least on average)*

# How to Identify Good PE Funds?



Pick past winners!

Returns to private equity are **highly persistent.** (Why?)

Issue: everyone wants to get in funds with strong performance

- Who gets access?
- What incentive effects does this create?

# PE Performance: Persistence



General Partner (PE fund) performance is persistent:

| | | | | | |
|---|---|---|---|---|---|
| $\text{PME}_{t-1}$ | 0.54 | 0.49 | | | 0.46 |
| | (0.17) | (0.21) | | | (0.21) |
| $\text{PME}_{t-2}$ | | 0.28 | 0.39 | | 0.28 |
| | | (0.13) | (0.14) | | (0.13) |
| $\text{PME}_{t-3}$ | | | | 0.32 | |
| | | | | (0.24) | |
| log(Size) | | | | | 0.09 |
| | | | | | (0.05) |
| log(Sequence) | | | | | 0.20 |
| | | | | | (0.20) |

# Private Equity: Other Issues



- Often the binding constraint for successful PE funds is **investment opportunities**, not investors
- Leads to the **money-chasing-deals** phenomenon
- During periods of high flows into private equity, funds pay higher valuations for their acquisitions

# Next Class: April 7



**Topics:** Hedge Funds I

- Hedge fund strategies
- Pairs trading
- Merger arbitrage